

Problem 18

Determine the limit to which the following expression tends as $n \rightarrow \infty$:

$$a_n = \frac{1}{n} \sum_{k=0}^n e^{-k}.$$

Solution

Step 1: Recognize the Geometric Series

The sum $\sum_{k=0}^n e^{-k}$ is a finite geometric series with:

- First term: $a = e^0 = 1$
- Common ratio: $r = e^{-1} = \frac{1}{e}$
- Number of terms: $n + 1$

Step 2: Apply the Geometric Series Formula

The formula for the sum of a finite geometric series is:

$$\sum_{k=0}^n ar^k = a \cdot \frac{1 - r^{n+1}}{1 - r}.$$

In our case:

$$\sum_{k=0}^n e^{-k} = \sum_{k=0}^n (e^{-1})^k = \frac{1 - (e^{-1})^{n+1}}{1 - e^{-1}} = \frac{1 - e^{-(n+1)}}{1 - e^{-1}}.$$

Step 3: Simplify the Denominator

The denominator can be rewritten as:

$$1 - e^{-1} = 1 - \frac{1}{e} = \frac{e - 1}{e}.$$

Therefore:

$$\sum_{k=0}^n e^{-k} = \frac{1 - e^{-(n+1)}}{\frac{e-1}{e}} = \frac{e(1 - e^{-(n+1)})}{e - 1}.$$

Step 4: Express a_n

Now we can write:

$$a_n = \frac{1}{n} \sum_{k=0}^n e^{-k} = \frac{1}{n} \cdot \frac{e(1 - e^{-(n+1)})}{e - 1} = \frac{e(1 - e^{-(n+1)})}{n(e - 1)}.$$

Step 5: Evaluate the Limit as $n \rightarrow \infty$

We need to find:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{e(1 - e^{-(n+1)})}{n(e - 1)}.$$

As $n \rightarrow \infty$, we have $e^{-(n+1)} \rightarrow 0$, so:

$$1 - e^{-(n+1)} \rightarrow 1.$$

Therefore:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{e \cdot 1}{n(e - 1)} = \lim_{n \rightarrow \infty} \frac{e}{n(e - 1)} = \frac{e}{e - 1} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = \frac{e}{e - 1} \cdot 0 = 0.$$

Step 6: Alternative Interpretation

We can also understand this result intuitively. The sum $\sum_{k=0}^n e^{-k}$ converges to a finite value as $n \rightarrow \infty$:

$$\sum_{k=0}^{\infty} e^{-k} = \frac{1}{1 - e^{-1}} = \frac{e}{e - 1}.$$

This is a constant (approximately 1.582). Therefore:

$$a_n = \frac{1}{n} \sum_{k=0}^n e^{-k} \approx \frac{1}{n} \cdot \frac{e}{e - 1} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

We're dividing a bounded sum by n , which grows without bound, so the ratio must approach zero.

Detailed Calculation

To be more precise:

$$a_n = \frac{e(1 - e^{-(n+1)})}{n(e - 1)} = \frac{e}{n(e - 1)} - \frac{e \cdot e^{-(n+1)}}{n(e - 1)} = \frac{e}{n(e - 1)} - \frac{e^{-n}}{n(e - 1)}.$$

Both terms approach zero as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{e}{n(e - 1)} = 0, \quad \lim_{n \rightarrow \infty} \frac{e^{-n}}{n(e - 1)} = 0.$$

Final Answer

The limit of the expression is:

$$\boxed{\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^n e^{-k} = 0.}$$